Data Science Job Posting

DATA CLEANING

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Business Question:

What is the impact of Job Titles on Salary Amount?

Our Focus

We were interested in looking at the relationship between Job Titles and Salary Amount to see if there was a difference in earning potential compared to job level

Dataset Overview

We noticed that most of the columns looked messy:

- The “Job Title” column had lots of similar positions given slightly different titles
- The “Salary Estimate” column contained unnecessary information such as “Glassdoor est.”
- We also noticed that there were some missing values

```
df = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/Uncleaned_DS_jobs.csv')
df.head()
```

	index	Job Title	Salary Estimate	Job Description	Rating	Company Name	Location	Headquarters	Size	Founded	Type of ownership	Industry	Sector	Revenue	Compet
0	0	Sr Data Scientist	$137K-$171K (Glassdoor est.)	Description\n\nThe Senior Data Scientist is re...	3.1	Healthfirst\n3.1	New York, NY	New York, NY	1001 to 5000 employees	1993	Nonprofit Organization	Insurance Carriers	Insurance	Unknown / Non-Applicable	Emblemh United Group,
1	1	Data Scientist	$137K-$171K (Glassdoor est.)	Secure our Nation, Ignite your Future\n\nJoin ...	4.2	ManTech\n4.2	Chantilly, VA	Herndon, VA	5001 to 10000 employees	1968	Company - Public	Research & Development	Business Services	$1 to $2 billion (USD)	
2	2	Data Scientist	$137K-$171K (Glassdoor est.)	Overview\n\n\nAnalysis Group is one of the lar...	3.8	Analysis Group\n3.8	Boston, MA	Boston, MA	1001 to 5000 employees	1981	Private Practice / Firm	Consulting	Business Services	$100 to $500 million (USD)	
3	3	Data Scientist	$137K-$171K (Glassdoor est.)	JOB DESCRIPTION:\n\nDo you have a passion for ...	3.5	INFICON\n3.5	Newton, MA	Bad Ragaz, Switzerland	501 to 1000 employees	2000	Company - Public	Electrical & Electronic Manufacturing	Manufacturing	$100 to $500 million (USD)	Instru F Va Agilent
4	4	Data Scientist	$137K-$171K (Glassdoor est.)	Data Scientist\nAffinity Solutions / Marketing...	2.9	Affinity Solutions\n2.9	New York, NY	New York, NY	51 to 200 employees	1998	Company - Private	Advertising & Marketing	Business Services	Unknown / Non-Applicable	Com S Carc

Cleaning Job Titles

```python
#cleaning of job titles

def job_title(title):
  title_lower = title.lower()
  if 'data scientist' in title_lower:
    return 'Data scientist'
  elif 'data engineer' in title_lower:
    return 'Data engineer'
  elif 'analyst' in title_lower:
    return 'Analyst'
  elif 'machine learning' in title_lower:
    return 'MLE'
  elif 'manager' in title_lower:
    return 'Manager'
  elif 'director' in title_lower:
    return 'Director'
  else:
    return 'na'

df['Updated Job Title'] = df['Job Title'].apply(job_title)
```

We first narrowed down the job title options by combining similar titles

```python
#List of the most common job types
df['Updated Job Title'].value_counts()
```

Updated Job Title	count
Data scientist	455
na	69
Analyst	55
Data engineer	47
MLE	36
Manager	7
Director	3

dtype: int64

We were then able to look at the most common job types

Updated Salaries

```
#removing 'Employer est.' and 'Glassdoor est.'
df['Salary Estimate'] = df['Salary Estimate'].apply(lambda x:x.replace('(Employer est.)','').replace(' (Glassdoor est.)','').replace('$',''))
sorted(df['Salary Estimate'].unique())
```

```
['101K-165K',
 '105K-167K',
 '110K-163K',
 '112K-116K',
 '122K-146K',
```

We removed the ‘Employer est.’, Glassdoor est.’, and ‘k’ to help the viewer focus on the dollar amount over the label of the company providing the data. This way, our data visualization is simpler.

```
# Remove letter K, and multiply values with 1000

def salary_estimate(values):
    df = values.replace('K','')
    data2 = df.split('-')
    data3 = [str(int(datas)*1000) for datas in data2]
    return '-'.join(data3)

df['Salary Estimate'] =  df['Salary Estimate'].apply(salary_estimate)

sorted(df['Salary Estimate'].unique())
```

```
['101000-165000',
 '105000-167000',
 '110000-163000',
 '112000-116000',
 '122000-146000',
 '124000-198000',
```

Adding New Columns

We added several new columns to our dataset so that we could include additional variables to display in our bar chart.

```
# Adding columna 'min_salary','max_salary', and 'avg_salary'

df['Min Salary'] = df['Salary Estimate'].apply(lambda x: int(x.split('-')[0]))
df['Max Salary'] = df['Salary Estimate'].apply(lambda x: int(x.split('-')[1]))
# Calculate average salary
df['Average Salary'] = (df['Min Salary'] + df['Max Salary']) / 2
df['Average Salary'] = df['Average Salary'].astype(int)
```

Cleaning Salaries

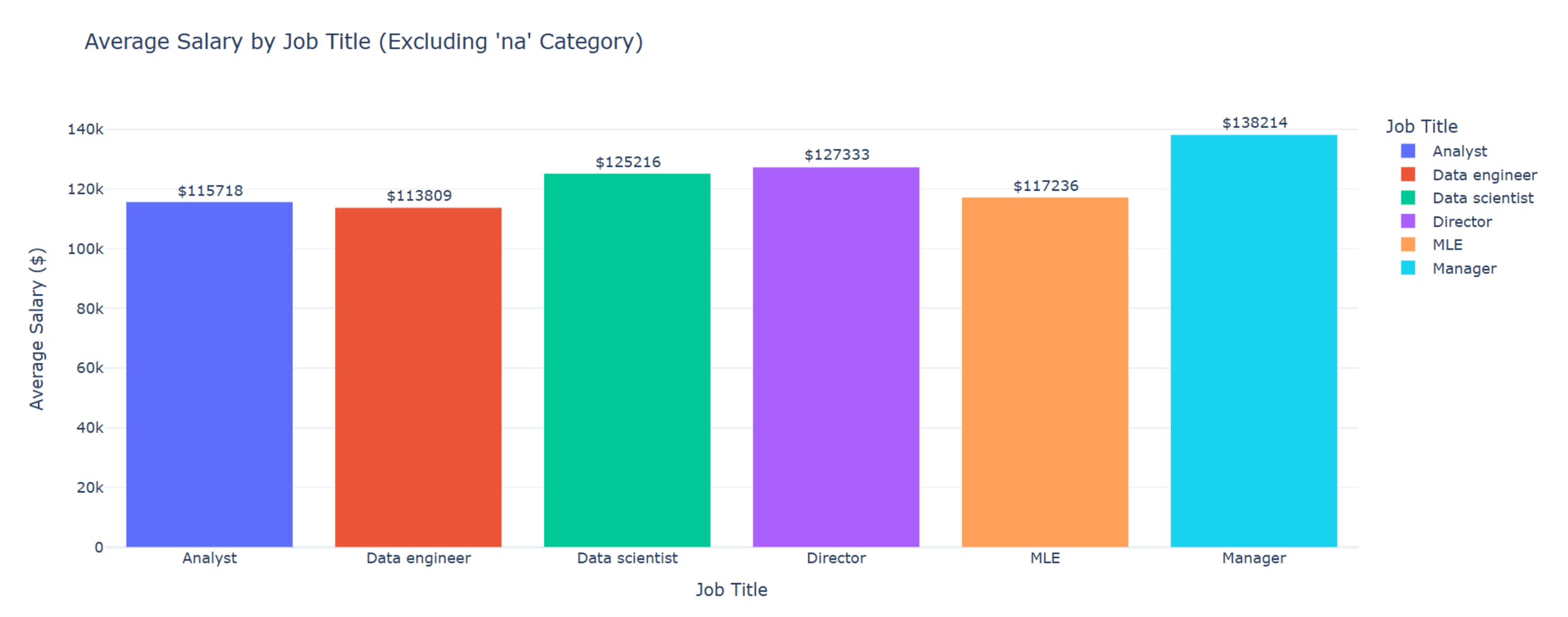
Before:

	index	Job Title	Salary Estimate
0	0	Sr Data Scientist	$137K-$171K (Glassdoor est.)
1	1	Data Scientist	$137K-$171K (Glassdoor est.)
2	2	Data Scientist	$137K-$171K (Glassdoor est.)
3	3	Data Scientist	$137K-$171K (Glassdoor est.)
4	4	Data Scientist	$137K-$171K (Glassdoor est.)

After:

	index	Job Title	Salary Estimate	Updated Job Title	Min Salary	Max Salary	Average Salary
0	0	Sr Data Scientist	137000-171000	Data scientist	137000	171000	154000
1	1	Data Scientist	137000-171000	Data scientist	137000	171000	154000
2	2	Data Scientist	137000-171000	Data scientist	137000	171000	154000
3	3	Data Scientist	137000-171000	Data scientist	137000	171000	154000
4	4	Data Scientist	137000-171000	Data scientist	137000	171000	154000

Visualization



Chart

What We Found

The salary varies little across all positions

Senior positions pay the most (Managers, Directors, etc.)

Data engineers and data analysts get paid the least

Machine Learning Engineers are in the middle

Summary

A good visualization requires cleaned-up data

Excluding the category ‘na’ and including a max salary positively affected our analysis

The bar chart is one of the best ways present job titles when compared to other elements

Thank You